E-commerce Data Analysis using SQL

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Project Type: SQL Data Analysis Project

1. Project Overview

This project analyzes an e-commerce dataset using SQL to understand sales performance, customer behavior, and product trends. The goal is to extract business insights that can help improve revenue, optimize customer retention, and support data-driven decision making.

2. Business Questions

1. What is the total revenue and total number of orders?

2. How does revenue change month by month?

3. What is the distribution of order values?

4. What is the order cancellation rate?

5. Who are the highest value customers?

6. What are the most popular product categories?

3. Overall Business Performance

This section calculates the overall performance of the business including total revenue, total orders, and average order value. These metrics provide a high-level view of the company's sales performance.

```
3.1 total orders

SELECT COUNT(DISTINCT order_id) AS total_orders

FROM olist_order_items_dataset;
```

Insight

The dataset contains approximately **99,441 orders**, indicating a large transaction volume for analysis.

total_orders
98666

3.2 total revenue

```
SELECT ROUND(SUM(price),2) AS total_revenue

FROM olist_order_items_dataset;
```

Insight

The total revenue generated from delivered orders is approximately **14.4 million**, indicating strong overall sales performance.

total_revenue
13650848.82

3.3 average order value

```
SELECT ROUND(SUM(oi.price)/COUNT(DISTINCT o.order_id),2) AS
avg_order_value

FROM olist_orders_dataset o

JOIN olist_order_items_dataset oi

ON o.order_id = oi.order_id;
```

Insight

The average order value is around **146**, suggesting that most purchases fall within a mid-price range.

avg_order_revenue
146.15

4. Monthly Revenue Analysis

This analysis examines how revenue changes over time. By grouping orders by month, we can identify seasonality patterns and growth trends.

```
SELECT DATE_FORMAT(order_purchase_timestamp,'%y-%m') AS month,

ROUND(SUM(oi.price),2) AS monthly_revenue
```

```
FROM olist_orders_dataset o

JOIN olist_order_items_dataset oi

ON o.order_id = oi.order_id

GROUP BY month

ORDER BY month;
```

month	monthly_revenue
16-09	399.75
16-10	52214.25
16-12	10.9
17-01	127785.73
17-02	260828.47
17-03	392540.45
17-04	379083.35
17-05	547444.44
17-06	453334.75
17-07	531705.38
17-08	606393.97
17-09	656219.08
17-10	698328.18
17-11	1077644.34
17-12	781550.35
18-01	1002781.64
18-02	891977.22
18-03	1044971.6
18-04	1055542.15

5. Order Value Distribution

This section analyzes the distribution of order values to understand purchasing behavior. It helps identify whether most orders are small purchases or high-value transactions.

```
SELECT o.order_id,

ROUND(SUM(oi.price),2) AS order_revenue

FROM olist_orders_dataset o

JOIN olist_order_items_dataset oi

ON o.order_id = oi.order_id
```

```
WHERE order_status = 'delivered'

GROUP BY o.order_id

ORDER BY order_revenue DESC;
```

order_id	order_revenue
03caa2c082116e1d31e67e9ae3700499	13440
f5136e38d1a14a4dbd87dff67da82701	12998
b239ca7cd485940b31882363b52e6674	8118
736e1922ae60d0d6a89247b851902527	7160
0812eb902a67711a1cb742b3cdaa65ae	6735
fefacc66af859508bf1a7934eab1e97f	6729
cb418bb4b6115438d20b0141ed5bc9e0	6396
1d54db601b417ccdfd3b7071955e6cc7	6210
0af28d87520565eb3b57c9b2abe1a2cc	5999.98
52e6988a13f9dd7d567b0816dba52a03	5999.78
2cc9089445046817a7539d90805e6e5a	5934.6
da8be3bb62e9bf01e2e1a3bfd74ebd1a	5380
a96610ab360d42a2e5335a3998b4718a	4799
e130d3f737127ae52681b9338aea9b64	4740
199af31afc78c699f0dbf71fb178d4d4	4690
8dbc85d1447242f3b127dda390d56e19	4590

6. Order Cancellation Analysis

This section measures the percentage of canceled orders. Understanding cancellation rates helps identify potential operational or customer experience issues.

```
SELECT

COUNT(order_id) AS total_orders,

SUM(CASE WHEN order_status ='canceled' THEN 1 ELSE 0 END) AS canceled_orders,

ROUND(SUM(CASE WHEN order_status ='canceled' THEN 1 ELSE 0
END)/COUNT(order_id)*100,2) AS canceled_rate

FROM olist_orders_dataset;
```

Insight

The cancellation rate is relatively low (around **0.63%**), but cancellations of high-value orders could still affect total revenue.

	total_orders	caceled_orders	canceled_rate
▶	105102	659	0.63

7. Customer Value Analysis

This analysis identifies high-value customers by calculating total spending per customer. It helps businesses focus on retaining their most valuable customers.

```
SELECT

      o.customer_id,

      ROUND(SUM(oi.price),2) AS customer_revenue

FROM olist_orders_dataset o

JOIN olist_order_items_dataset oi

ON o.order_id = oi.order_id

WHERE o.order_status = 'delivered'

GROUP BY o.customer_id

ORDER BY customer_revenue DESC;
```

	customer_id	customer_revenue
▶	1617b1357756262bfa56ab541c47bc16	13440
	3fd6777bbce08a352fddd04e4a7cc8f6	12998
	e9b0d0eb3015ef1c9ce6cf5b9dcbee9f	8118
	ec5b2ba62e574342386871631fafd3fc	7160
	c6e2731c5b391845f6800c97401a43a9	6735
	f48d464a0baaea338cb25f816991ab1f	6729
	a7ea318cbe9df2ec79ab37cd7ca2135d	6396
	7e7d2271d6f55b03b0bd4615c5f2de6e	6210
	0a209a88c2e3dc2981c79ad85c558059	5999.98
	8dd4c93bfbebe2a77657d46ef959a7ac	5999.78
	05455dfa7cd02f13d132aa7a6a9729c6	5934.6
	3f083b9f62e687be8d84684c7d54c8db	5380
	df55c14d1476a9a3467f131269c2477f	4799
	[illegible]	4740

8. Customer Behavior Analysis

This section studies how often customers place orders and whether they return for repeat purchases.

```
SELECT o.customer_id,

       COUNT(DISTINCT o.order_id) AS order_count

FROM olist_orders_dataset o

WHERE o.order_status = 'delivered'

GROUP BY o.customer_id;
```

customer_id	order_count
00012a2ce6f8dcda20d059ce98491703	1
000161a058600d5901f007fab4c27140	1
0001fd6190edaaf884bcaf3d49edf079	1
0002414f95344307404f0ace7a26f1d5	1
000379cdec625522490c315e70c7a9fb	1
0004164d20a9e969af783496f3408652	1
000419c5494106c306a97b5635748086	1

```
WITH customer_stats AS(

SELECT o.customer_id,

       COUNT(DISTINCT o.order_id) AS order_count,

       ROUND(SUM(oi.price),2) AS total_revenue

FROM olist_orders_dataset o

JOIN olist_order_items_dataset oi

ON o.order_id = oi.order_id

WHERE o.order_status = 'delivered'

GROUP BY o.customer_id

),

Ranked_customers AS(

SELECT order_count,

       total_revenue,

       customer_id,
```

```
        NTILE(10)OVER(ORDER BY total_revenue DESC) AS revenue_group
FROM customer_stats
)
SELECT revenue_group,
        COUNT(customer_id) AS customer_count,
        ROUND(AVG(order_count),2) AS avg_order_count,
        ROUND(SUM(total_revenue),2) AS group_revenue
        FROM ranked_customers
        GROUP BY revenue_group
        ORDER BY revenue_group DESC;
```

revenue_group	customer_count	avg_order_count	revenue_group
1	9648	1.00	5844399.31
2	9648	1.00	2180365.91
3	9648	1.00	1540285.43
4	9648	1.00	1201053.43
5	9648	1.00	960379.64
6	9648	1.00	763983.25
7	9648	1.00	589891.32
8	9648	1.00	452826.96
9	9647	1.00	317725.52
10	9647	1.00	181008.5

```
WITH customer_stats AS(
SELECT o.customer_id,
        COUNT(DISTINCT o.order_id) AS order_count,
        ROUND(SUM(oi.price),2) AS total_revenue
FROM olist_orders_dataset o
JOIN olist_order_items_dataset oi
ON o.order_id = oi.order_id
WHERE o.order_status = 'delivered'
GROUP BY o.customer_id
```

```
)

SELECT CASE

      WHEN order_count>=3 THEN 'High_frequency' ELSE 'Low_frequency' END AS
frequency_group,

      COUNT(customer_id) AS customer_count,

      SUM(total_revenue) AS total_revenue,

      ROUND(SUM(total_revenue)/SUM(SUM(total_revenue))OVER()*100,2) AS
revenue_percent

      FROM customer_stats

         GROUP BY

      CASE

      WHEN order_count>=3 THEN 'High_frequency' ELSE 'Low_frequency' END ;
```

frequency_group	customer_count	total_revenue	revenue_percent
Lowfrequency	96478	14031919.270012075	100

9. Product Category Analysis

This analysis identifies the best-performing product categories based on revenue and order volume.

```
WITH category_stats AS(

SELECT

      p.product_category_name,

      COUNT(DISTINCT o.order_id) AS orders_count,

      SUM(oi.price) AS total_revenue

FROM olist_orders_dataset o

JOIN olist_order_items_dataset oi

ON o.order_id = oi.order_id

JOIN olist_products_dataset p

ON p.product_id = oi.product_id

WHERE o.order_status = 'delivered'
```

```
GROUP BY p.product_category_name

)

SELECT

    product_category_name,

    orders_count,

    ROUND(total_revenue,2) AS total_revenue,

    ROUND(total_revenue/orders_count,2) AS avg_order_value

    FROM category_stats

    ORDER BY total_revenue DESC;
```

Insight

The category **beleza_saude** has the highest number of orders, indicating strong and stable market demand. This category could be considered a core product segment for the business.

product_category_name	orders_count	total_revenue	avg_order_value
beleza_saude	8647	1614478.78	186.71
relogios_presentes	5495	1582002.11	287.9
cama_mesa_banho	9272	1415276.09	152.64
esporte_lazer	7530	1269458.73	168.59
informatica_acessorios	6530	1217222.47	186.4
moveis_decoracao	6307	949235.51	150.51
cool_stuff	3559	832369.88	233.88
automotivo	3810	808951.98	212.32
utilidades_domesticas	5743	795180.96	138.46
ferramentas_jardim	3448	661055.03	191.72
brinquedos	3804	649355.33	170.7
bebes	2808	573823.33	204.35
perfumaria	3086	532167.24	172.45
telefonia	4093	423509.69	103.47
moveis_escritorio	1254	349914.53	279.04
papelaria	2264	310362.94	137.09

Result 38

10. Key Business Insights

1. A small percentage of orders contribute a large portion of revenue.

2. Customer repeat purchase behavior appears limited.

3. Some product categories significantly outperform others.

4. Monitoring high-value order cancellations is important for revenue stability.

5.The top 10% of customers contribute approximately 41.65% of total revenue, indicating a moderate level of customer concentration.

11. Business Recommendations

1. Introduce loyalty programs to improve repeat purchases.

2. Focus marketing efforts on high-value customers.

3. Promote top-performing product categories.

4. Investigate causes of order cancellations to reduce revenue loss.

12.Project Summary

This project analyzed an e-commerce dataset using SQL to understand revenue structure, customer behavior, and product performance.

The analysis found that a small percentage of orders contribute a large portion of revenue, while customer repeat purchase behavior remains relatively low.

Product category analysis also revealed that certain categories generate significantly higher demand.

These insights can help businesses improve marketing strategies, customer retention, and product investment decisions.